Inserting Fmap with Coercive Subtyping

This paper shows how to use subtyping to infer uses of the basic monad operations.

CCS Concepts: • **Software and its engineering → Functional languages**.

Additional Key Words and Phrases: subtyping, functors

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1 Introduction

Typeclass abstractions like Functor, Applicative, and Monad are essential to modern functional programming. The liberal application of type class combinators allows concise programming even in the presence of complex effects and data structures. For example, consider writing a Haskell program that increments integers wrapped in a `Maybe` type.

```
incrMaybe : Maybe Int -> Maybe Int
incrMaybe (Just x) = Just (succ x)
incrMaybe Nothing  = Nothing
```

This naive definition of `incrMaybe` has to unwrap its argument with a pattern match and then wrap its output back up. A more idiomatic definition uses <$>, the fmap combinator, which handles the unwrapping and wrapping.

```
incrMaybe x = succ <$> x
```

In the presence of sufficient type information, we are interested in further reducing the boilerplate involved by making fmap implicit. This permits writing `incrMaybe` with no use of fmap at all.

```
incrMaybe x = succ x
```

It seems as if there is only one canonical way to make this program make sense, and automatically lifting functions to operate on arrays and other data structures is already prevalent both in programming languages and mathematical notation.

In this paper, we study a *coercive subtyping* discipline that infers where to insert implicit fmaps using subtyping. Each derivation of $A <: B$ then corresponds to a coercion $A \to B$ expressed in a simple combinator calculus. In coercive subtyping, type inference is coupled to an elaboration pass, which inserts coercions in the program based on the results of subtyping queries. This makes the operational meaning of surface programs depend on the result of type inference, which breaks with a more naive view of inference as merely tacking missing type annotations on to a program with already fixed semantics. Indeed, many programming languages confound their users with strange behavior involving implicit coercions, whether they are inserted statically or dynamically.

We can avoid these confusions by establishing *coherence*, a basic metatheoretic property of systems of coercions. This problem was studied as early as 1980 by Reynolds [2], who outlined an

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$$A ::= \mathbb{B} \mid A \to A \mid F\,A$$

$$\textsc{Refl}\ \frac{}{A \le A} \qquad \textsc{Trans}\ \frac{A \le B \qquad B \le C}{A \le C} \qquad \textsc{Arr}\ \frac{U \le A \qquad B \le V}{A \to B \le U \to V}$$

$$\textsc{Mono}\ \frac{A \le B}{F\,A \le F\,B} \qquad \textsc{Fmap}\ \frac{}{A \to B \le F\,A \to F\,B}$$

Fig. 1. Types and subtype relationships for a calculus with implicit fmap.

algebraic approach to proving the coherence of coercions. In his view, coercions form a *category*: there are types $A, B, \ldots$ and between any two types there is a set of coercions $\mathrm{Coe}(A, B)$. There are always identity coercions $\mathsf{id}_A \in \mathrm{Coe}(A, A)$, which do nothing, and there are associative and unital composition operations $\circ : \mathrm{Coe}(B, C) \times \mathrm{Coe}(A, B) \to \mathrm{Coe}(A, C)$. The coercions are expected to form a preorder, that is, for every pair of arrows $c1, c2 \in \mathrm{Coe}(A, B)$, $c1 \equiv c2$. He also studies their interaction with generic operators of the language, but we will restrict our attention to the coercions themselves.

As is typical in programming languages, we establish our coercions are coherent by showing every coercion $c : A \le B$ can be put into a normal form $\mathsf{nf}\,c$ that is equivalent to c, and then show that normal forms are unique. Rather than relying on rewriting, we adopt the technique of *normalization by evaluation* (NbE).

Besides giving some confidence in the safety of this programming model, the proof of soundness has independent interest as a case study in the application of normalization by evaluation to implicit coercions.

2 Overview

The types and subtyping rules at issue are shown in Figure 3. We call these the declarative rules (as opposed to the algorithmic rules). The only nonstandard rule is Fmap, which permits using functions $A \to B$ at the lifted type $F\,A \to F\,B$. This rule may be iterated, permitting the use of functions at $F\,A \to F\,B$, $F\,F\,A \to F\,F\,B$, and so on. These rules serve as a good specification of the system, but they have practical and theoretical limitations.

The declarative rules are not suitable for an implementation because they are not deterministic: too many rules apply to possible subtyping queries. Besides the fact that it is always possible to appeal to Trans, queries of the form $A \to B \le F\,U \to F\,V$ must choose between Arr and Fmap.

They are also questionable from a semantic point of view because a single judgment may have many derivations and the coercions associated with these derivations may not be equivalent.

[todo: Describe the design of normalized coercions by constructing “multiplication tables”]

3 Coercions

To define coercions, we introduce proof terms to the subtyping judgment and then define an equivalence on proof terms. In the new judgment $c : A \le B$, the proof term c is an intrinsically typed coercion; for proof rules that consume multiple derivations the coercion c is defined using a coercion combinator (Fig. 2).

Definition 1 (Coercion congruence). *The equivalence relation $\equiv$ is the least congruence $\equiv$ such that for all $c : A \leq B$,*

$$\begin{aligned}
\mathsf{id} \cdot c &\equiv c \\
c \cdot \mathsf{id} &\equiv c \\
c_1 \cdot (c_2 \cdot c_3) &\equiv (c_1 \cdot c_2) \cdot c_3 \\
\mathsf{mono}\ (\mathsf{id}_A) &\equiv \mathsf{id}_{F\,A} \\
\mathsf{mono}\ (c_1 \cdot c_2) &\equiv \mathsf{mono}\ c_1 \cdot \mathsf{mono}\ c_2 \\
\mathsf{arr}\ \mathsf{id}_A\ \mathsf{id}_B &\equiv \mathsf{id}_{A \to B} \\
\mathsf{arr}\ (c_1 \cdot c_2)\ (c_3 \cdot c_4) &\equiv \mathsf{arr}\ c_2\ c_3 \cdot \mathsf{arr}\ c_1\ c_4 \\
\mathsf{fmap} \cdot \mathsf{arr}\ (\mathsf{mono}\ c_1)\ (\mathsf{mono}\ c_2) &\equiv \mathsf{arr}\ c_1\ c_2 \cdot \mathsf{fmap}
\end{aligned}$$

In order, these laws say that coercions form a category, mono is a functor on that category, and arr is a bifunctor on that category. The final law about fmap is more complicated becuase fmap is not a functor on coercions but instead a functor on types. In the equational theory of coercions, all that can be said about is how it interacts with other coercions on arrow types.

3.1 Compiling coercions correctly

In order to justify the equational theory and show that these coercion combinators can be soundly compiled to a programming language, we provide a translation into typed lambda calculus equipped with an abstract functor F. Then we prove that the translation preserves equivalence, which is given by the equational theory in the source language and by $\beta\eta$-equivalence and fmap laws in the target.

$$\begin{aligned}
\mathsf{C}(\mathbb{B}) &= \mathsf{bool} \\
\mathsf{C}(A \to B) &= \mathsf{C}(A) \texttt{->} \mathsf{C}(B) \\
\mathsf{C}(F\,A) &= \mathsf{F}\mathsf{C}(A)
\end{aligned}$$

$$\begin{aligned}
\mathsf{C}(\mathsf{id}_A) &= \lambda x : \mathsf{C}(A).\, x \\
\mathsf{C}(c_1 \cdot c_2) &= \lambda x.\, \mathsf{C}(c_2)\ (c_1\ x) \\
\mathsf{C}(\mathsf{arr}\ c_1\ c_2) &= \lambda f\, x.\, c_2\ (f\ (c_1\ x)) \\
\mathsf{C}(\mathsf{mono}\ c) &= \mathsf{fmap}\ \mathsf{C}(c) \\
\mathsf{C}(\mathsf{fmap}) &= \mathsf{fmap}
\end{aligned}$$

Proposition 2 (C sound [TODO: agda icon]). *For all $c_1, c_2 : A \leq B$,*

$$c_1 \equiv c_2 \implies \mathsf{C}(c_1) \equiv_{\beta\eta\mathsf{F}} \mathsf{C}(c_2)$$

3.2 Normal forms for coercions

Proposition 3 (Uniqueness of normal forms [TODO: agda icon]). *Let A, B, U, and V be types.*

- *If $n : A \leq_{\mathsf{n}} B$ and $n' : A \leq_{\mathsf{n}} B$ then $n = n'$.*
- *If $na : A, B \leq_{\mathsf{na}} U, V$ and $na' : A, B \leq_{\mathsf{na}} U, V$ then $na = na'$.*

4 Coherence

- The equational theory of coercions: congruence closure of category laws plus functor/bifunctor laws and the "special fmap law"
  - informal justification in terms of fmap laws and associativity of composition (for terminating programs.... if we are going to be careful...)
  - explanation of why this is necessary at all
- The equational theory of normalized coercions: just syntactic equality.
- The reify function

$\boxed{c : A \leq B}$

$$\textsc{CId}\ \frac{}{\mathsf{id}_A : A \leq_{\mathsf{n}} A} \qquad \textsc{CComp}\ \frac{c_1 : A \leq B \qquad c_2 : B \leq C}{c_1 \cdot c_2 : A \leq C}$$

$$\textsc{CArr}\ \frac{c_1 : U \leq A \qquad c_2 : B \leq V}{\mathsf{arr}\ c_1\ c_2 : A \to B \leq U \to V}$$

$$\textsc{CMono}\ \frac{c : A \leq B}{\mathsf{mono}\ c : F\,A \leq F\,B} \qquad \textsc{CFmap}\ \frac{}{\mathsf{fmap} : A \to B \leq F\,A \to F\,B}$$

Fig. 2. Declarative coercions.

$\boxed{n : A \leq_{\mathsf{n}} B}$ $\boxed{na : A, B \leq_{\mathsf{na}} U, V}$

$$\textsc{NId}\ \frac{}{\mathsf{nid} : \mathbb{B} \leq_{\mathsf{n}} \mathbb{B}} \qquad \textsc{NMono}\ \frac{n : A \leq_{\mathsf{n}} B}{\mathsf{nmono}\ n : F\,A \leq_{\mathsf{n}} F\,B}$$

$$\textsc{NArr}\ \frac{na : A, B \leq_{\mathsf{na}} U, V}{\mathsf{narr}\ na : A \to B \leq_{\mathsf{n}} F\,A \to F\,B} \qquad \textsc{NAFmap}\ \frac{na : A, B \leq_{\mathsf{na}} U, V}{na \cdot \mathsf{nfmap} : A, B \leq_{\mathsf{na}} F\,U, F\,V}$$

$$\textsc{NAArr}\ \frac{n_1 : U \leq_{\mathsf{n}} A \qquad n_2 : B \leq_{\mathsf{n}} V}{\mathsf{naarr}\ n_1\ n_2 : A, B \leq_{\mathsf{na}} U, V}$$

Fig. 3. Normalized coercions.

- The evaluation procedure (more involved, establishes admissibility of transitivity and reflexivity)

Proposition 4 (Soundness of evaluation). *If* $c_1 \equiv c_2$, *then* $[\![c_1]\!] = [\![c_2]\!]$.

Proposition 5 (Reify inverts evaluation). *If* $c : A \leq B$, *then* $\mathsf{reify}\ [\![c]\!] \equiv c$.

- normalization is reify-eval
- normalization is idempotent
- normalization is sound
- normalization is complete

5 Implementation in Agda

The type .. is quotiented by a congruence relation.

The model justifying the congruence relation interprets types in an Agda universe. While effectful, potentially nonterminating programming languages have different equational theories than Agda, the equational theory of coercions should be stable.

6 Perspectives

Appealing extensions: applicatives, monads, …

Difficulties in doing that: loss of the peel invariant, etc

Appealing results: coherence for an entire lambda calculus

Difficulties in doing that: difficult to set up an untyped semantics in order to prove a bracketing theorem

7 Related Work

Reynolds [2].

Coherence proof for monoidal categories [1].

8 Conclusion

References

[1] Ilya Beylin and Peter Dybjer. 1995. Extracting a proof of coherence for monoidal categories from a proof of normalization for monoids. In *TYPES (Lecture Notes in Computer Science)*. Springer, 47–61.

[2] John C. Reynolds. 1980. Using category theory to design implicit conversions and generic operators. In *Semantics-Directed Compiler Generation (Lecture Notes in Computer Science)*. Springer, 211–258.